

TUTONCHANY, Russian Federation

WMO: 235890

Lat: 64.208N

Lon: 93.793E

Elev: 106

StdP: 100.06

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -52.9	(c) -49.6	(d) -54.6	(e) 0.0	(f) -52.5	(g) -51.9	(h) 0.0	(i) -49.6	(j) 6.7	(k) -17.7	(l) 5.6	(m) -18.6	(n) 0.1	(o) 90	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	28.9	17.4	26.8	16.6	24.4	15.9	19.2	25.1	17.9	23.5	16.9	22.4	2.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	12.3	21.4	15.9	11.4	20.1	14.7	10.6	19.0	55.1	24.9	51.0	23.6	47.8	22.2	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.1	6.0	5.2	-52.2	31.9	4.3	0.9	-55.3	32.6	-57.8	33.1	-60.2	33.6	-63.4	34.3		
			-51.9	20.9	4.4	1.3	-55.1	21.9	-57.7	22.7	-60.2	23.4	-63.4	24.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-6.8	-32.5	-25.5	-13.6	-4.4	2.9	13.9	16.2	12.7	4.7	-5.1	-24.3	-27.5
	DBStd	18.46	11.92	9.64	8.97	5.90	4.55	4.85	3.32	3.85	4.21	6.97	10.61	11.29
	HDD10.0	6563	1319	993	731	432	224	18	1	19	166	469	1029	1162
	HDD18.3	9193	1577	1226	989	682	479	146	82	177	409	727	1279	1420
	CDD10.0	443	0	0	0	0	4	136	193	103	7	0	0	0
	CDD18.3	31	0	0	0	0	0	13	16	2	0	0	0	0
	CDH23.3	662	0	0	0	0	1	298	307	54	1	0	0	0
	CDH26.7	174	0	0	0	0	0	91	78	6	0	0	0	0
Wind	WSAvg	2.0	1.4	1.5	2.1	2.8	2.8	2.3	2.0	1.8	2.1	2.1	1.5	1.5
Precipitation	PrecAvg	476	27	21	19	24	40	59	54	62	49	53	38	31
	PrecMax	658	52	33	47	44	62	99	154	121	80	84	65	56
	PrecMin	298	4	10	5	8	15	9	12	25	21	12	14	9
	PrecStd	85	10	7	11	10	13	27	34	26	19	18	13	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.8	-0.8	4.6	10.6	21.9	31.6	30.8	27.5	21.1	9.3	-0.1	-2.5
		MCWB	-8.0	-2.4	0.7	4.2	12.1	17.3	19.0	17.3	12.6	6.1	-1.1	-3.4
	2%	DB	-9.4	-6.2	2.7	8.6	17.4	29.4	29.0	24.6	16.3	6.2	-3.3	-4.6
		MCWB	-10.2	-7.2	0.2	3.0	9.6	16.8	18.1	16.3	10.9	4.1	-4.0	-5.4
	5%	DB	-11.5	-8.7	0.8	6.4	13.5	27.2	27.0	22.7	13.4	4.5	-4.8	-7.2
		MCWB	-12.2	-9.4	-1.4	2.1	7.3	16.1	17.2	15.7	9.1	3.0	-5.4	-8.0
	10%	DB	-14.2	-11.0	-0.8	4.2	10.5	24.7	24.8	20.3	11.2	2.7	-8.1	-10.9
		MCWB	-14.8	-11.7	-2.5	1.0	5.5	15.2	16.5	14.8	8.3	1.4	-8.8	-11.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.9	-2.4	1.8	5.1	12.8	19.9	21.6	19.2	14.6	6.4	-1.2	-3.4
		MCDB	-7.0	-0.9	3.2	8.8	21.4	28.0	28.0	24.7	19.4	9.3	-0.1	-2.8
	2%	WB	-10.1	-7.1	0.2	3.7	10.1	18.2	20.0	17.7	11.5	4.7	-3.8	-5.4
		MCDB	-9.5	-6.5	2.5	7.3	17.0	25.6	25.8	22.1	14.9	5.8	-3.2	-4.6
	5%	WB	-12.1	-9.4	-1.1	2.5	7.9	17.1	18.6	16.5	9.8	3.0	-5.5	-7.7
		MCDB	-11.5	-8.7	0.7	5.7	12.5	24.6	23.8	20.9	12.8	4.1	-4.8	-7.0
	10%	WB	-14.7	-11.7	-2.5	1.4	5.9	16.0	17.5	15.4	8.5	1.5	-8.8	-11.5
		MCDB	-14.2	-11.0	-0.8	4.2	9.8	22.7	22.6	19.7	10.8	2.6	-8.1	-11.0
Mean Daily Temperature Range	5% DB	MDBR	8.7	10.4	13.6	11.7	9.9	14.0	13.1	11.4	8.8	6.9	9.0	8.8
		MCDBR	9.1	7.9	11.9	13.9	15.6	19.1	19.1	16.9	13.0	6.4	10.5	9.7
		MCWBR	8.8	7.5	9.8	9.3	8.9	8.0	8.7	9.0	7.7	4.8	10.0	9.1
	5% WB	MCDBR	9.0	7.9	11.0	11.8	14.8	16.0	14.3	13.8	11.1	5.6	10.6	9.0
		MCWBR	8.7	7.4	9.3	8.4	9.1	7.4	7.5	8.0	7.1	4.5	10.1	8.6
Clear-Sky Solar Irradiance	taub		0.230	0.281	0.314	0.342	0.361	0.376	0.427	0.407	0.340	0.287	0.210	0.215
	taud		1.866	1.980	2.015	2.042	2.169	2.399	2.238	2.262	2.408	2.179	1.898	1.863
	Ebn at Noon		285	589	744	817	841	841	775	752	742	596	308	97
	Edh at Noon		30	71	105	129	126	103	118	105	74	57	27	12
All-Sky Solar Radiation	RadAvg		0.15	0.86	2.26	4.06	5.09	5.56	4.89	3.27	1.95	0.79	0.26	0.06
	RadStd		0.02	0.09	0.24	0.42	0.38	0.55	0.45	0.38	0.28	0.14	0.04	0.01

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page